

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question
If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

- Answer**
- A. -7 and -3
 - B. -2 and 2
 - C. 2 and 7
 - D. 8 and 13

Correct Answer: B

Rationale
Choice B is correct. Multiplying both sides of the given equation by $(3)(13)$, or 39 , yields $(39)\left(\frac{x+6}{3}\right) = (39)\left(\frac{x+6}{13}\right)$, or $13(x + 6) = 3(x + 6)$. Subtracting $3(x + 6)$ from both sides of this equation yields $10(x + 6) = 0$. Dividing both sides of this equation by 10 yields $x + 6 = 0$. Therefore, if $\frac{x+6}{3} = \frac{x+6}{13}$, then the value of $x + 6$ is 0 . It follows that of the given choices, the value of $x + 6$ is between -2 and 2 .
Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.